

Solução do Problema grow22 - Solver HiGHS

Análise Computacional

6 de outubro de 2025

1 Informações do Problema

Informações do Problema:

- Nome: grow22
- Número de Variáveis: 645
- Número de Restrições: 300
- Inviabilidade Primal: 0.000e+00
- Inviabilidade Dual: 5.995e-12
- Valor Primal: -1.069e+08
- Valor Dual: 0.000e+00
- Gap: -5.871e-24
- Número de Iterações: 20

2 Variáveis Primais e Custos Reduzidos

Tabela 1: Variáveis primais (x) e custos reduzidos (z) do problema grow22

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
1	2.001e+05	1	0.000e+00
2	4.163e+03	2	0.000e+00
3	7.845e+03	3	0.000e+00
4	4.362e+03	4	0.000e+00
5	1.401e+04	5	0.000e+00
6	2.680e+04	6	0.000e+00
7	9.182e+03	7	0.000e+00
8	1.180e+04	8	0.000e+00
9	6.291e+04	9	-6.153e+01
10	1.174e+04	10	0.000e+00
11	1.379e+04	11	0.000e+00
12	2.016e+04	12	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
13	7.315e+04	13	0.000e+00
14	7.934e+04	14	0.000e+00
15	6.820e+02	15	0.000e+00
16	5.767e+03	16	0.000e+00
17	1.723e+04	17	0.000e+00
18	2.985e+04	18	0.000e+00
19	1.093e+04	19	0.000e+00
20	7.593e+05	20	0.000e+00
21	0.000e+00	21	4.230e+00
22	0.000e+00	22	3.000e+00
23	0.000e+00	23	6.906e-01
24	1.234e+05	24	0.000e+00
25	9.638e+02	25	0.000e+00
26	0.000e+00	26	-9.548e-15
27	0.000e+00	27	-4.086e-14
28	0.000e+00	28	1.465e-14
29	0.000e+00	29	1.155e-14
30	0.000e+00	30	0.000e+00
31	0.000e+00	31	-1.243e-14
32	2.699e+04	32	0.000e+00
33	0.000e+00	33	-7.994e-15
34	0.000e+00	34	-4.423e-13
35	6.204e+03	35	0.000e+00
36	4.775e+04	36	0.000e+00
37	0.000e+00	37	-4.352e-14
38	0.000e+00	38	0.000e+00
39	0.000e+00	39	0.000e+00
40	0.000e+00	40	0.000e+00
41	1.658e+04	41	-4.539e-13
42	2.658e+03	42	0.000e+00
43	5.524e+05	43	-6.439e-15
44	1.005e+06	44	0.000e+00
45	0.000e+00	45	2.309e-14
46	1.322e+06	46	0.000e+00
47	5.162e+03	47	0.000e+00
48	4.551e+04	48	0.000e+00
49	8.727e+04	49	0.000e+00
50	1.736e+04	50	0.000e+00
51	1.125e+04	51	0.000e+00
52	6.291e+04	52	-6.153e+01
53	3.404e+04	53	-4.441e-16
54	2.151e+04	54	0.000e+00
55	0.000e+00	55	2.043e-14
56	0.000e+00	56	-4.707e-14
57	1.155e+05	57	0.000e+00
58	3.025e+03	58	0.000e+00
59	9.904e+03	59	0.000e+00
60	3.190e+04	60	0.000e+00
61	0.000e+00	61	-3.977e-13

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
62	2.264e+04	62	3.275e-15
63	3.550e+04	63	0.000e+00
64	7.500e+05	64	-1.770e+00
65	0.000e+00	65	4.086e-14
66	1.000e+06	66	-1.309e+00
67	3.448e+04	67	0.000e+00
68	0.000e+00	68	-1.599e-14
69	2.655e+05	69	0.000e+00
70	0.000e+00	70	0.000e+00
71	0.000e+00	71	-4.619e-14
72	0.000e+00	72	-2.842e-14
73	3.096e+03	73	0.000e+00
74	0.000e+00	74	-1.599e-14
75	1.153e+04	75	0.000e+00
76	8.634e+03	76	0.000e+00
77	0.000e+00	77	4.143e-13
78	0.000e+00	78	0.000e+00
79	4.396e+04	79	0.000e+00
80	0.000e+00	80	-1.750e-13
81	2.960e+03	81	0.000e+00
82	8.666e+03	82	3.146e-13
83	0.000e+00	83	-3.286e-14
84	0.000e+00	84	5.955e-13
85	1.132e+04	85	0.000e+00
86	1.788e+05	86	0.000e+00
87	0.000e+00	87	3.553e-15
88	6.037e+05	88	0.000e+00
89	7.688e+05	89	0.000e+00
90	7.499e+03	90	0.000e+00
91	3.195e+04	91	0.000e+00
92	7.780e+04	92	0.000e+00
93	6.666e+04	93	4.441e-15
94	2.369e+04	94	0.000e+00
95	6.291e+04	95	-6.153e+01
96	2.644e+04	96	0.000e+00
97	4.118e+04	97	-4.441e-15
98	2.411e+04	98	0.000e+00
99	0.000e+00	99	-4.396e-14
100	2.474e+05	100	0.000e+00
101	0.000e+00	101	2.420e-14
102	0.000e+00	102	3.011e-13
103	5.411e+04	103	0.000e+00
104	3.316e+04	104	1.308e-13
105	2.415e+03	105	0.000e+00
106	1.510e+05	106	0.000e+00
107	0.000e+00	107	4.230e+00
108	5.000e+05	108	-3.000e+00
109	1.000e+06	109	-3.309e+00
110	0.000e+00	110	-7.105e-15

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
111	3.891e+03	111	0.000e+00
112	0.000e+00	112	0.000e+00
113	0.000e+00	113	0.000e+00
114	0.000e+00	114	0.000e+00
115	5.946e+04	115	0.000e+00
116	3.333e+04	116	2.132e-14
117	4.738e+03	117	0.000e+00
118	8.948e+03	118	0.000e+00
119	1.702e+04	119	-3.997e-15
120	3.032e+03	120	0.000e+00
121	2.112e+03	121	0.000e+00
122	3.867e+04	122	0.000e+00
123	8.939e+04	123	0.000e+00
124	2.782e+03	124	0.000e+00
125	7.188e+03	125	0.000e+00
126	2.582e+04	126	0.000e+00
127	0.000e+00	127	-1.492e-13
128	5.814e+03	128	0.000e+00
129	0.000e+00	129	-3.997e-15
130	1.102e+06	130	0.000e+00
131	0.000e+00	131	3.997e-15
132	1.032e+06	132	0.000e+00
133	3.900e+03	133	0.000e+00
134	4.166e+04	134	0.000e+00
135	2.474e+04	135	0.000e+00
136	4.802e+03	136	0.000e+00
137	0.000e+00	137	1.776e-15
138	6.291e+04	138	-6.153e+01
139	7.109e+03	139	0.000e+00
140	6.648e+03	140	0.000e+00
141	0.000e+00	141	4.441e-15
142	0.000e+00	142	-4.352e-14
143	0.000e+00	143	-1.670e-13
144	0.000e+00	144	2.598e-14
145	0.000e+00	145	3.024e-13
146	0.000e+00	146	-2.842e-14
147	1.773e+04	147	0.000e+00
148	0.000e+00	148	9.437e-16
149	4.006e+05	149	0.000e+00
150	7.500e+05	150	-1.770e+00
151	0.000e+00	151	5.684e-14
152	1.000e+06	152	-5.309e+00
153	0.000e+00	153	9.770e-15
154	2.373e+03	154	0.000e+00
155	0.000e+00	155	1.465e-14
156	0.000e+00	156	-4.796e-13
157	0.000e+00	157	6.839e-14
158	0.000e+00	158	1.599e-14
159	2.964e+04	159	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
160	0.000e+00	160	1.474e-13
161	0.000e+00	161	8.527e-14
162	0.000e+00	162	1.688e-14
163	0.000e+00	163	0.000e+00
164	0.000e+00	164	-5.862e-14
165	3.563e+04	165	0.000e+00
166	0.000e+00	166	2.212e-13
167	2.431e+03	167	0.000e+00
168	4.009e+03	168	0.000e+00
169	0.000e+00	169	4.108e-14
170	0.000e+00	170	8.660e-15
171	0.000e+00	171	5.440e-15
172	0.000e+00	172	1.465e-14
173	4.720e+04	173	0.000e+00
174	0.000e+00	174	-1.688e-14
175	1.045e+06	175	0.000e+00
176	5.597e+03	176	0.000e+00
177	3.839e+04	177	0.000e+00
178	1.207e+04	178	0.000e+00
179	1.478e+04	179	0.000e+00
180	2.467e+04	180	0.000e+00
181	6.291e+04	181	-6.153e+01
182	2.924e+04	182	0.000e+00
183	2.846e+04	183	0.000e+00
184	1.672e+04	184	0.000e+00
185	0.000e+00	185	-8.527e-14
186	2.342e+05	186	0.000e+00
187	0.000e+00	187	1.710e-14
188	0.000e+00	188	2.951e-13
189	1.941e+04	189	0.000e+00
190	1.951e+04	190	0.000e+00
191	1.150e+04	191	0.000e+00
192	5.293e+05	192	0.000e+00
193	0.000e+00	193	4.230e+00
194	0.000e+00	194	3.000e+00
195	1.000e+06	195	-3.309e+00
196	2.122e+04	196	0.000e+00
197	0.000e+00	197	4.263e-14
198	0.000e+00	198	0.000e+00
199	0.000e+00	199	5.258e-13
200	0.000e+00	200	1.599e-14
201	0.000e+00	201	-2.105e-13
202	3.333e+04	202	3.004e-12
203	6.281e+03	203	0.000e+00
204	1.409e+04	204	0.000e+00
205	1.702e+04	205	0.000e+00
206	0.000e+00	206	4.086e-14
207	0.000e+00	207	8.438e-14
208	2.954e+04	208	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
209	9.811e+04	209	0.000e+00
210	1.949e+03	210	0.000e+00
211	0.000e+00	211	-3.118e-13
212	0.000e+00	212	-1.465e-14
213	0.000e+00	213	2.220e-15
214	0.000e+00	214	0.000e+00
215	1.836e+05	215	0.000e+00
216	1.116e+06	216	0.000e+00
217	1.319e+05	217	0.000e+00
218	1.054e+06	218	0.000e+00
219	4.815e+03	219	0.000e+00
220	4.289e+04	220	0.000e+00
221	1.189e+05	221	-1.812e-13
222	0.000e+00	222	1.736e-12
223	0.000e+00	223	4.707e-14
224	6.291e+04	224	-6.153e+01
225	1.003e+04	225	0.000e+00
226	1.310e+04	226	0.000e+00
227	4.527e+03	227	0.000e+00
228	0.000e+00	228	-1.403e-13
229	0.000e+00	229	-2.753e-14
230	0.000e+00	230	1.332e-14
231	2.170e+03	231	0.000e+00
232	3.337e+04	232	0.000e+00
233	3.316e+04	233	-2.776e-15
234	9.949e+03	234	0.000e+00
235	2.285e+05	235	0.000e+00
236	7.500e+05	236	-1.770e+00
237	0.000e+00	237	1.421e-14
238	1.000e+06	238	-1.309e+00
239	0.000e+00	239	2.220e-14
240	1.082e+05	240	0.000e+00
241	1.809e+04	241	0.000e+00
242	0.000e+00	242	0.000e+00
243	0.000e+00	243	-6.661e-14
244	1.698e+04	244	0.000e+00
245	2.592e+04	245	0.000e+00
246	0.000e+00	246	-5.995e-12
247	0.000e+00	247	3.411e-13
248	0.000e+00	248	6.670e-13
249	0.000e+00	249	4.619e-14
250	0.000e+00	250	3.997e-14
251	2.662e+04	251	0.000e+00
252	0.000e+00	252	-1.519e-13
253	1.722e+03	253	0.000e+00
254	0.000e+00	254	2.243e-14
255	0.000e+00	255	7.350e-14
256	1.048e+04	256	0.000e+00
257	0.000e+00	257	-8.216e-15

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
258	0.000e+00	258	-3.708e-14
259	1.726e+05	259	0.000e+00
260	4.677e+05	260	0.000e+00
261	0.000e+00	261	7.550e-15
262	9.092e+03	262	4.441e-14
263	5.811e+04	263	0.000e+00
264	4.832e+04	264	0.000e+00
265	0.000e+00	265	1.770e-12
266	3.072e+04	266	-4.308e-12
267	6.291e+04	267	-6.153e+01
268	3.404e+04	268	6.546e-13
269	1.778e+04	269	0.000e+00
270	2.948e+04	270	3.109e-14
271	0.000e+00	271	-1.092e-13
272	9.354e+04	272	0.000e+00
273	0.000e+00	273	1.421e-14
274	7.549e+03	274	0.000e+00
275	2.884e+04	275	0.000e+00
276	8.481e+03	276	0.000e+00
277	1.117e+04	277	0.000e+00
278	1.748e+05	278	0.000e+00
279	0.000e+00	279	4.230e+00
280	5.000e+05	280	-3.000e+00
281	0.000e+00	281	6.906e-01
282	1.002e+05	282	0.000e+00
283	0.000e+00	283	1.998e-14
284	0.000e+00	284	6.439e-15
285	2.320e+03	285	0.000e+00
286	4.853e+04	286	0.000e+00
287	4.114e+04	287	0.000e+00
288	1.917e+04	288	0.000e+00
289	1.536e+04	289	5.970e-12
290	1.616e+04	290	0.000e+00
291	1.702e+04	291	-6.519e-13
292	0.000e+00	292	-1.066e-14
293	1.065e+04	293	0.000e+00
294	2.334e+04	294	0.000e+00
295	0.000e+00	295	2.087e-13
296	1.544e+03	296	0.000e+00
297	5.746e+03	297	0.000e+00
298	7.982e+03	298	0.000e+00
299	0.000e+00	299	5.551e-15
300	4.372e+03	300	0.000e+00
301	0.000e+00	301	4.130e-14
302	1.017e+06	302	0.000e+00
303	3.638e+04	303	0.000e+00
304	1.038e+06	304	0.000e+00
305	3.406e+03	305	0.000e+00
306	0.000e+00	306	-9.281e-14

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
307	1.189e+05	307	-1.958e-13
308	0.000e+00	308	1.706e-12
309	1.194e+04	309	0.000e+00
310	6.291e+04	310	-6.153e+01
311	1.002e+04	311	0.000e+00
312	2.085e+04	312	0.000e+00
313	1.651e+04	313	0.000e+00
314	0.000e+00	314	-1.630e-13
315	1.310e+05	315	0.000e+00
316	0.000e+00	316	7.550e-15
317	0.000e+00	317	1.710e-14
318	6.193e+04	318	3.819e-14
319	2.433e+04	319	0.000e+00
320	2.264e+04	320	-5.940e-15
321	7.605e+05	321	0.000e+00
322	7.500e+05	322	-1.770e+00
323	0.000e+00	323	-5.773e-15
324	1.000e+06	324	-1.309e+00
325	6.390e+03	325	0.000e+00
326	2.810e+04	326	0.000e+00
327	0.000e+00	327	1.044e-14
328	0.000e+00	328	0.000e+00
329	0.000e+00	329	1.057e-13
330	5.946e+04	330	2.514e-13
331	1.114e+04	331	0.000e+00
332	1.172e+04	332	0.000e+00
333	0.000e+00	333	4.832e-13
334	0.000e+00	334	1.776e-14
335	0.000e+00	335	0.000e+00
336	1.474e+04	336	0.000e+00
337	1.612e+04	337	0.000e+00
338	0.000e+00	338	2.398e-14
339	8.259e+02	339	0.000e+00
340	0.000e+00	340	-1.044e-14
341	3.097e+04	341	-3.597e-14
342	0.000e+00	342	7.327e-15
343	1.132e+04	343	1.421e-14
344	2.640e+05	344	0.000e+00
345	0.000e+00	345	-3.331e-14
346	0.000e+00	346	-1.688e-14
347	1.043e+06	347	0.000e+00
348	6.244e+03	348	0.000e+00
349	1.119e+05	349	0.000e+00
350	2.307e+03	350	0.000e+00
351	0.000e+00	351	1.668e-12
352	0.000e+00	352	-3.464e-14
353	6.291e+04	353	-6.153e+01
354	2.742e+04	354	0.000e+00
355	3.647e+04	355	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
356	7.474e+03	356	0.000e+00
357	0.000e+00	357	-1.785e-13
358	3.301e+05	358	0.000e+00
359	0.000e+00	359	3.109e-15
360	3.506e+02	360	0.000e+00
361	1.177e+04	361	0.000e+00
362	1.463e+04	362	0.000e+00
363	9.565e+03	363	0.000e+00
364	0.000e+00	364	-7.216e-15
365	0.000e+00	365	4.230e+00
366	0.000e+00	366	3.000e+00
367	1.000e+06	367	-3.309e+00
368	0.000e+00	368	5.240e-14
369	2.592e+04	369	0.000e+00
370	0.000e+00	370	0.000e+00
371	0.000e+00	371	0.000e+00
372	8.062e+04	372	0.000e+00
373	5.946e+04	373	2.398e-14
374	6.485e+03	374	0.000e+00
375	0.000e+00	375	4.796e-14
376	1.691e+04	376	0.000e+00
377	1.702e+04	377	2.665e-15
378	0.000e+00	378	0.000e+00
379	4.203e+03	379	0.000e+00
380	1.113e+04	380	0.000e+00
381	1.778e+05	381	0.000e+00
382	8.142e+02	382	0.000e+00
383	0.000e+00	383	0.000e+00
384	2.599e+04	384	0.000e+00
385	0.000e+00	385	-1.554e-15
386	1.132e+04	386	0.000e+00
387	0.000e+00	387	9.770e-15
388	1.108e+06	388	0.000e+00
389	0.000e+00	389	-1.821e-14
390	1.032e+06	390	0.000e+00
391	3.704e+03	391	0.000e+00
392	0.000e+00	392	-1.776e-15
393	3.013e+04	393	0.000e+00
394	0.000e+00	394	1.665e-12
395	1.157e+04	395	0.000e+00
396	6.291e+04	396	-6.153e+01
397	7.223e+03	397	0.000e+00
398	9.593e+03	398	0.000e+00
399	0.000e+00	399	-1.332e-15
400	0.000e+00	400	-1.794e-13
401	0.000e+00	401	-4.441e-15
402	2.763e+02	402	0.000e+00
403	1.002e+04	403	0.000e+00
404	0.000e+00	404	-1.776e-15

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
405	3.316e+04	405	-1.887e-15
406	1.560e+03	406	0.000e+00
407	3.964e+05	407	0.000e+00
408	7.500e+05	408	-1.770e+00
409	0.000e+00	409	-5.773e-15
410	1.000e+06	410	-5.309e+00
411	2.989e+03	411	0.000e+00
412	2.337e+04	412	0.000e+00
413	0.000e+00	413	0.000e+00
414	0.000e+00	414	0.000e+00
415	3.819e+04	415	0.000e+00
416	3.611e+03	416	0.000e+00
417	0.000e+00	417	-2.874e-12
418	3.691e+03	418	0.000e+00
419	7.678e+03	419	0.000e+00
420	0.000e+00	420	-1.332e-15
421	0.000e+00	421	0.000e+00
422	3.105e+03	422	0.000e+00
423	7.733e+03	423	0.000e+00
424	9.933e+04	424	0.000e+00
425	7.423e+02	425	0.000e+00
426	6.392e+03	426	0.000e+00
427	0.000e+00	427	-2.220e-14
428	1.342e+04	428	0.000e+00
429	5.779e+03	429	0.000e+00
430	0.000e+00	430	-4.219e-15
431	0.000e+00	431	1.066e-14
432	5.691e+05	432	0.000e+00
433	1.036e+06	433	0.000e+00
434	2.899e+03	434	0.000e+00
435	0.000e+00	435	1.021e-14
436	0.000e+00	436	4.885e-15
437	2.428e+04	437	0.000e+00
438	0.000e+00	438	3.286e-14
439	6.291e+04	439	-6.153e+01
440	1.487e+04	440	0.000e+00
441	1.929e+04	441	0.000e+00
442	0.000e+00	442	3.997e-15
443	0.000e+00	443	-1.297e-13
444	0.000e+00	444	1.599e-14
445	0.000e+00	445	5.107e-15
446	0.000e+00	446	4.885e-15
447	1.548e+04	447	0.000e+00
448	1.720e+04	448	0.000e+00
449	0.000e+00	449	1.388e-15
450	3.395e+05	450	0.000e+00
451	0.000e+00	451	4.230e+00
452	5.000e+05	452	-3.000e+00
453	1.000e+06	453	-3.309e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
454	0.000e+00	454	-2.220e-14
455	0.000e+00	455	0.000e+00
456	0.000e+00	456	-3.997e-15
457	0.000e+00	457	0.000e+00
458	0.000e+00	458	-2.176e-14
459	0.000e+00	459	-2.043e-14
460	8.027e+03	460	0.000e+00
461	0.000e+00	461	-5.329e-14
462	1.891e+04	462	0.000e+00
463	0.000e+00	463	-5.773e-15
464	0.000e+00	464	0.000e+00
465	0.000e+00	465	1.474e-13
466	5.691e+03	466	0.000e+00
467	0.000e+00	467	-3.553e-14
468	4.319e+02	468	0.000e+00
469	3.633e+03	469	0.000e+00
470	0.000e+00	470	-7.105e-15
471	0.000e+00	471	-3.775e-15
472	0.000e+00	472	0.000e+00
473	0.000e+00	473	-1.354e-14
474	1.143e+06	474	0.000e+00
475	6.074e+03	475	0.000e+00
476	1.044e+06	476	0.000e+00
477	6.358e+03	477	0.000e+00
478	4.507e+04	478	0.000e+00
479	1.036e+05	479	0.000e+00
480	0.000e+00	480	1.155e-14
481	1.488e+04	481	0.000e+00
482	6.291e+04	482	-6.153e+01
483	2.845e+04	483	0.000e+00
484	2.097e+04	484	0.000e+00
485	1.136e+04	485	0.000e+00
486	0.000e+00	486	-1.248e-13
487	1.305e+05	487	0.000e+00
488	0.000e+00	488	1.021e-14
489	0.000e+00	489	8.438e-15
490	3.722e+04	490	0.000e+00
491	2.094e+04	491	0.000e+00
492	1.118e+04	492	0.000e+00
493	4.569e+05	493	0.000e+00
494	7.500e+05	494	-1.770e+00
495	2.839e+03	495	0.000e+00
496	1.000e+06	496	-1.309e+00
497	0.000e+00	497	2.136e+00
498	0.000e+00	498	1.634e+00
499	0.000e+00	499	6.906e-01
500	0.000e+00	500	2.511e+00
501	0.000e+00	501	1.341e+00
502	0.000e+00	502	1.698e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
503	0.000e+00	503	5.072e+00
504	0.000e+00	504	3.566e+00
505	0.000e+00	505	3.500e+01
506	0.000e+00	506	1.374e+00
507	0.000e+00	507	1.353e+00
508	0.000e+00	508	1.994e+00
509	0.000e+00	509	2.092e+00
510	0.000e+00	510	2.273e+00
511	0.000e+00	511	5.064e-01
512	0.000e+00	512	4.315e-01
513	0.000e+00	513	5.819e-01
514	0.000e+00	514	4.396e-01
515	0.000e+00	515	2.684e-01
516	0.000e+00	516	4.545e-01
517	6.443e+03	517	0.000e+00
518	1.599e+05	518	0.000e+00
519	8.135e+04	519	0.000e+00
520	4.244e+03	520	0.000e+00
521	7.170e+03	521	0.000e+00
522	2.919e+03	522	0.000e+00
523	2.526e+04	523	0.000e+00
524	7.566e+03	524	0.000e+00
525	6.291e+04	525	-3.640e+01
526	7.903e+03	526	0.000e+00
527	1.321e+04	527	0.000e+00
528	2.170e+04	528	0.000e+00
529	1.148e+04	529	0.000e+00
530	1.008e+05	530	0.000e+00
531	5.905e+02	531	0.000e+00
532	8.568e+02	532	0.000e+00
533	4.408e+04	533	0.000e+00
534	1.439e+04	534	0.000e+00
535	3.910e+03	535	0.000e+00
536	9.483e+04	536	0.000e+00
537	0.000e+00	537	2.094e+00
538	0.000e+00	538	1.366e+00
539	7.454e+04	539	0.000e+00
540	0.000e+00	540	-5.773e-15
541	1.329e+05	541	0.000e+00
542	0.000e+00	542	0.000e+00
543	0.000e+00	543	1.830e-13
544	0.000e+00	544	-3.775e-14
545	0.000e+00	545	-1.328e-13
546	1.065e+04	546	0.000e+00
547	0.000e+00	547	-1.776e-14
548	3.145e+04	548	-2.984e-13
549	0.000e+00	549	9.932e-13
550	0.000e+00	550	1.246e-13
551	7.999e+03	551	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
552	5.595e+03	552	0.000e+00
553	4.842e+04	553	0.000e+00
554	4.694e+02	554	0.000e+00
555	0.000e+00	555	5.218e-14
556	3.097e+04	556	3.186e-14
557	4.080e+03	557	0.000e+00
558	0.000e+00	558	0.000e+00
559	0.000e+00	559	2.109e-15
560	1.144e+06	560	0.000e+00
561	4.405e+05	561	0.000e+00
562	1.047e+06	562	0.000e+00
563	6.332e+03	563	0.000e+00
564	4.934e+04	564	0.000e+00
565	1.044e+05	565	0.000e+00
566	0.000e+00	566	1.865e-14
567	1.478e+04	567	0.000e+00
568	6.291e+04	568	-3.640e+01
569	3.404e+04	569	9.386e-13
570	2.690e+04	570	0.000e+00
571	0.000e+00	571	-9.992e-15
572	0.000e+00	572	-2.220e-15
573	1.130e+05	573	0.000e+00
574	0.000e+00	574	-1.632e-14
575	4.394e+03	575	0.000e+00
576	7.622e+03	576	0.000e+00
577	3.316e+04	577	-1.432e-14
578	1.227e+04	578	0.000e+00
579	5.273e+05	579	0.000e+00
580	7.500e+05	580	-3.906e+00
581	5.000e+05	581	-1.634e+00
582	1.000e+06	582	-2.000e+00
583	0.000e+00	583	2.094e+00
584	0.000e+00	584	1.588e+00
585	0.000e+00	585	6.726e-01
586	0.000e+00	586	2.443e+00
587	0.000e+00	587	1.312e+00
588	0.000e+00	588	1.662e+00
589	0.000e+00	589	4.971e+00
590	0.000e+00	590	3.492e+00
591	0.000e+00	591	3.435e+01
592	0.000e+00	592	1.342e+00
593	0.000e+00	593	1.314e+00
594	0.000e+00	594	1.940e+00
595	0.000e+00	595	2.047e+00
596	0.000e+00	596	2.224e+00
597	0.000e+00	597	4.934e-01
598	0.000e+00	598	4.212e-01
599	0.000e+00	599	5.676e-01
600	2.702e+03	600	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
601	0.000e+00	601	2.601e-01
602	0.000e+00	602	4.411e-01
603	6.428e+04	603	0.000e+00
604	5.987e+05	604	0.000e+00
605	1.040e+06	605	0.000e+00
606	6.246e+03	606	0.000e+00
607	4.162e+04	607	0.000e+00
608	1.164e+04	608	0.000e+00
609	1.390e+04	609	0.000e+00
610	1.234e+04	610	0.000e+00
611	6.291e+04	611	-1.172e+01
612	1.899e+04	612	0.000e+00
613	3.349e+04	613	0.000e+00
614	1.636e+04	614	0.000e+00
615	6.697e+03	615	0.000e+00
616	1.573e+05	616	0.000e+00
617	3.736e+02	617	0.000e+00
618	3.418e+03	618	0.000e+00
619	2.385e+04	619	0.000e+00
620	3.316e+04	620	-3.774e-01
621	1.156e+04	621	0.000e+00
622	4.016e+05	622	0.000e+00
623	3.504e+04	623	0.000e+00
624	5.000e+05	624	-6.222e+00
625	1.000e+06	625	-4.673e+00
626	0.000e+00	626	1.000e+00
627	0.000e+00	627	7.783e-01
628	0.000e+00	628	3.274e-01
629	0.000e+00	629	1.193e+00
630	0.000e+00	630	6.303e-01
631	0.000e+00	631	7.972e-01
632	0.000e+00	632	2.373e+00
633	0.000e+00	633	1.673e+00
634	0.000e+00	634	1.633e+01
635	0.000e+00	635	6.472e-01
636	0.000e+00	636	6.454e-01
637	0.000e+00	637	9.476e-01
638	0.000e+00	638	9.831e-01
639	0.000e+00	639	1.067e+00
640	0.000e+00	640	2.398e-01
641	0.000e+00	641	2.037e-01
642	0.000e+00	642	2.751e-01
643	0.000e+00	643	6.366e-01
644	0.000e+00	644	1.286e-01
645	0.000e+00	645	2.171e-01

3 Variáveis Duais (Multiplicadores de Lagrange)

Tabela 2: Variáveis duais (y) do problema grow22

Coordenada y	Valor y (Dual)
1	5.230e+00
2	4.000e+00
3	1.691e+00
4	6.147e+00
5	3.284e+00
6	4.158e+00
7	1.242e+01
8	8.731e+00
9	8.567e+01
10	3.364e+00
11	3.312e+00
12	4.882e+00
13	5.122e+00
14	5.564e+00
15	1.240e+00
16	1.056e+00
17	1.425e+00
18	1.076e+00
19	6.571e-01
20	1.113e+00
21	5.230e+00
22	4.000e+00
23	1.691e+00
24	6.147e+00
25	3.284e+00
26	4.158e+00
27	1.242e+01
28	8.731e+00
29	8.567e+01
30	3.364e+00
31	3.312e+00
32	4.882e+00
33	5.122e+00
34	5.564e+00
35	1.240e+00
36	1.056e+00
37	1.425e+00
38	1.076e+00
39	6.571e-01
40	1.113e+00
41	5.230e+00
42	4.000e+00
43	1.691e+00
44	6.147e+00
45	3.284e+00
46	4.158e+00
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Coordenada y	Valor y (Dual)
47	1.242e+01
48	8.731e+00
49	8.567e+01
50	3.364e+00
51	3.312e+00
52	4.882e+00
53	5.122e+00
54	5.564e+00
55	1.240e+00
56	1.056e+00
57	1.425e+00
58	1.076e+00
59	6.571e-01
60	1.113e+00
61	5.230e+00
62	4.000e+00
63	1.691e+00
64	6.147e+00
65	3.284e+00
66	4.158e+00
67	1.242e+01
68	8.731e+00
69	8.567e+01
70	3.364e+00
71	3.312e+00
72	4.882e+00
73	5.122e+00
74	5.564e+00
75	1.240e+00
76	1.056e+00
77	1.425e+00
78	1.076e+00
79	6.571e-01
80	1.113e+00
81	5.230e+00
82	4.000e+00
83	1.691e+00
84	6.147e+00
85	3.284e+00
86	4.158e+00
87	1.242e+01
88	8.731e+00
89	8.567e+01
90	3.364e+00
91	3.312e+00
92	4.882e+00
93	5.122e+00
94	5.564e+00
95	1.240e+00
Continua na próxima página	

Coordenada y	Valor y (Dual)
96	1.056e+00
97	1.425e+00
98	1.076e+00
99	6.571e-01
100	1.113e+00
101	5.230e+00
102	4.000e+00
103	1.691e+00
104	6.147e+00
105	3.284e+00
106	4.158e+00
107	1.242e+01
108	8.731e+00
109	8.567e+01
110	3.364e+00
111	3.312e+00
112	4.882e+00
113	5.122e+00
114	5.564e+00
115	1.240e+00
116	1.056e+00
117	1.425e+00
118	1.076e+00
119	6.571e-01
120	1.113e+00
121	5.230e+00
122	4.000e+00
123	1.691e+00
124	6.147e+00
125	3.284e+00
126	4.158e+00
127	1.242e+01
128	8.731e+00
129	8.567e+01
130	3.364e+00
131	3.312e+00
132	4.882e+00
133	5.122e+00
134	5.564e+00
135	1.240e+00
136	1.056e+00
137	1.425e+00
138	1.076e+00
139	6.571e-01
140	1.113e+00
141	5.230e+00
142	4.000e+00
143	1.691e+00
144	6.147e+00
Continua na próxima página	

Coordenada y	Valor y (Dual)
145	3.284e+00
146	4.158e+00
147	1.242e+01
148	8.731e+00
149	8.567e+01
150	3.364e+00
151	3.312e+00
152	4.882e+00
153	5.122e+00
154	5.564e+00
155	1.240e+00
156	1.056e+00
157	1.425e+00
158	1.076e+00
159	6.571e-01
160	1.113e+00
161	5.230e+00
162	4.000e+00
163	1.691e+00
164	6.147e+00
165	3.284e+00
166	4.158e+00
167	1.242e+01
168	8.731e+00
169	8.567e+01
170	3.364e+00
171	3.312e+00
172	4.882e+00
173	5.122e+00
174	5.564e+00
175	1.240e+00
176	1.056e+00
177	1.425e+00
178	1.076e+00
179	6.571e-01
180	1.113e+00
181	5.230e+00
182	4.000e+00
183	1.691e+00
184	6.147e+00
185	3.284e+00
186	4.158e+00
187	1.242e+01
188	8.731e+00
189	8.567e+01
190	3.364e+00
191	3.312e+00
192	4.882e+00
193	5.122e+00
Continua na próxima página	

Coordenada y	Valor y (Dual)
194	5.564e+00
195	1.240e+00
196	1.056e+00
197	1.425e+00
198	1.076e+00
199	6.571e-01
200	1.113e+00
201	5.230e+00
202	4.000e+00
203	1.691e+00
204	6.147e+00
205	3.284e+00
206	4.158e+00
207	1.242e+01
208	8.731e+00
209	8.567e+01
210	3.364e+00
211	3.312e+00
212	4.882e+00
213	5.122e+00
214	5.564e+00
215	1.240e+00
216	1.056e+00
217	1.425e+00
218	1.076e+00
219	6.571e-01
220	1.113e+00
221	5.230e+00
222	4.000e+00
223	1.691e+00
224	6.147e+00
225	3.284e+00
226	4.158e+00
227	1.242e+01
228	8.731e+00
229	8.567e+01
230	3.364e+00
231	3.312e+00
232	4.882e+00
233	5.122e+00
234	5.564e+00
235	1.240e+00
236	1.056e+00
237	1.425e+00
238	1.076e+00
239	6.571e-01
240	1.113e+00
241	3.094e+00
242	2.366e+00
Continua na próxima página	

Coordenada y	Valor y (Dual)
243	1.000e+00
244	3.636e+00
245	1.942e+00
246	2.459e+00
247	7.344e+00
248	5.165e+00
249	5.068e+01
250	1.990e+00
251	1.959e+00
252	2.888e+00
253	3.030e+00
254	3.292e+00
255	7.333e-01
256	6.249e-01
257	8.427e-01
258	6.366e-01
259	3.887e-01
260	6.582e-01
261	3.094e+00
262	2.366e+00
263	1.000e+00
264	3.636e+00
265	1.942e+00
266	2.459e+00
267	7.344e+00
268	5.165e+00
269	5.068e+01
270	1.990e+00
271	1.959e+00
272	2.888e+00
273	3.030e+00
274	3.292e+00
275	7.333e-01
276	6.249e-01
277	8.427e-01
278	6.366e-01
279	3.887e-01
280	6.582e-01
281	1.000e+00
282	7.783e-01
283	3.274e-01
284	1.193e+00
285	6.303e-01
286	7.972e-01
287	2.373e+00
288	1.673e+00
289	1.633e+01
290	6.472e-01
291	6.454e-01
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Coordenada y	Valor y (Dual)
292	9.476e-01
293	9.831e-01
294	1.067e+00
295	2.398e-01
296	2.037e-01
297	2.751e-01
298	6.366e-01
299	1.286e-01
300	2.171e-01